Arbiter: From PDF to Verdict — Formal Verification Stipulations for Multi-Agent Research Claim Debate

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Abstract

Research papers routinely overreach beyond their evidence. The replication crisis (36% of landmark results replicate), tripling retraction rates, and growing skepticism toward LLM-as-judge evaluation demand tools that distinguish what a paper *proves* from what it *claims*. We present Arbiter, an open-source framework (11K LOC Python, 7 LLM providers) that implements a complete PDF-to-verdict pipeline: it reads academic papers with formal or quantitative claims, extracts those claims, attempts machine verification via Knuckledragger (Z3) and SymPy, stages multi-agent debates with argumentative validity enforcement, and produces verdicts from a multi-provider evaluation panel. In an extended case study on *The AI Layoff Trap* (Falk & Tsoukalas, 2026), Arbiter assembled 9 specialist agents across 3 frontier providers (GPT-5.4, Claude Opus 4.6, Grok 4.20), which produced 141 substantive argument exchanges across 54 turns over 6 rounds ("argument hits" counts distinct claims advanced or rebutted per turn, not correctness). The debate surfaced 17 Proponent concessions (the Proponent agent argues for the paper's claims)—revealing that the paper's core theorem ($\alpha_{\mathrm{NE}} > \alpha_{\mathrm{CO}}$ when $\eta < 1, N > 1$) is mathematically correct, but its headline policy claims ("only a Pigouvian tax can fix over-automation") extend far beyond the mathematics. In an ablation study on the same paper, debate with verification elicits more concessions and catches more overreach than either component alone; we treat this as preliminary evidence pending evaluation across more papers. As a byproduct of building the verification backend, we characterize failure modes in LLM-generated Z3 proofs and report certificate validity rates for our own pipeline; we find that Z3-based proof generation requires careful audit, as LLMs frequently produce encodings that are technically satisfiable but do not faithfully represent the mathematical claim. We report these findings transparently; the verification backend is a work in progress, and honest measurement is a prerequisite to improvement. Code and all experimental artifacts are available at https://github.com/vishk23/arbiter. All experiments were conducted in April 2026 using frontier models available at that time (GPT-5.4, Claude Opus 4.6, Grok 4.20).

1 Introduction

The scientific literature is in crisis. Only 36% of landmark psychology results replicated in the Open Science Collaboration's massive 2015 effort [1], and subsequent meta-analyses across medicine, economics, and machine learning have found similar or worse rates. Retraction rates have tripled over the past decade [2], and a 2026 Nature survey found that 70% of researchers have tried and failed to reproduce another scientist's experiments [3]. The problem is not merely fraud: most non-replicating papers contain claims that *technically follow* from toy models or narrow experimental conditions but are presented as general truths. The gap between what a paper proves and what it claims is where research overreach lives.

Automated evaluation tools have emerged but fall short. Claim verification systems like SciFact [4] and PROClaim [5] check whether claims are supported by cited evidence but do not evaluate

the *logical structure* of arguments or the gap between formal results and informal conclusions. LLM-as-judge approaches—now ubiquitous in NeurIPS reviewing [9] and automated benchmarks—face mounting criticism. The PAJAMA framework and "Time to Impeach LLM-as-a-Judge" [11] demonstrated systematic biases: position bias, verbosity bias, self-enhancement bias, and susceptibility to adversarial prompt injection. Multi-agent debate [13] was proposed as a remedy, but recent theoretical work shows that under certain conditions debate reduces to a martingale—iterating does not improve accuracy beyond a single strong judge [12].

We argue that the missing ingredient is *formal verification as a constraint on debate*. When mathematical claims are machine-verified before debate begins, agents cannot dispute settled results; they must instead argue about the *gap* between verified theorems and unverified conclusions. This transforms debate from a persuasion game into a structured audit.

Arbiter implements this insight as a complete pipeline:

1. **Agentic Initialization.** Given a PDF, Arbiter extracts assumptions, propositions, equations, and policy claims, then selects specialist debate agents and providers automatically.
2. **Formally Verified Stipulations.** Knuckledragger (Z3) and SymPy verify quantitative claims, producing proof certificates that become debate stipulations—facts no agent may contest.
3. **Validity-Enforced Debate.** A layered gate system rejects fallacious, circular, or unsupported arguments before they enter the debate transcript. Agents that fail the gate must rewrite.
4. **Multi-Provider Evaluation Panel.** Three frontier models from different providers independently score the debate, reducing single-model bias. Providers are correlated but not identical; cross-provider disagreement is itself a signal.

Contributions.

- A PDF-to-verdict pipeline that combines formal verification stipulations with multi-agent debate for research claim evaluation—to our knowledge, no prior open system integrates both components.
- An extended case study on a published economics paper [21] in which Arbiter elicits 17 Proponent-agent concessions (the Proponent agent argues for the paper's position; concessions reflect positions the paper's own logic cannot defend under adversarial scrutiny), surfacing a gap between the formal model and the headline policy claims.
- Ablation evidence on the same paper that debate with verification elicits more overreach concessions than debate alone or a single LLM-as-judge call. A broader multi-paper evaluation is ongoing.
- An open-source system (11K LOC) supporting 7 LLM providers, a web dashboard, and multiple export formats.

2 Related Work

Claim Verification. SciFact [4] introduced claim verification against scientific abstracts. PROClaim [5] extends this to full-text reasoning chains with provenance tracking. Loki [6] and SciClaimEval [7] focus on claim extraction and classification. These systems verify whether a claim is *supported by cited evidence* but do not evaluate whether formal results *entail* the paper's broader conclusions—precisely the overreach gap Arbiter targets.

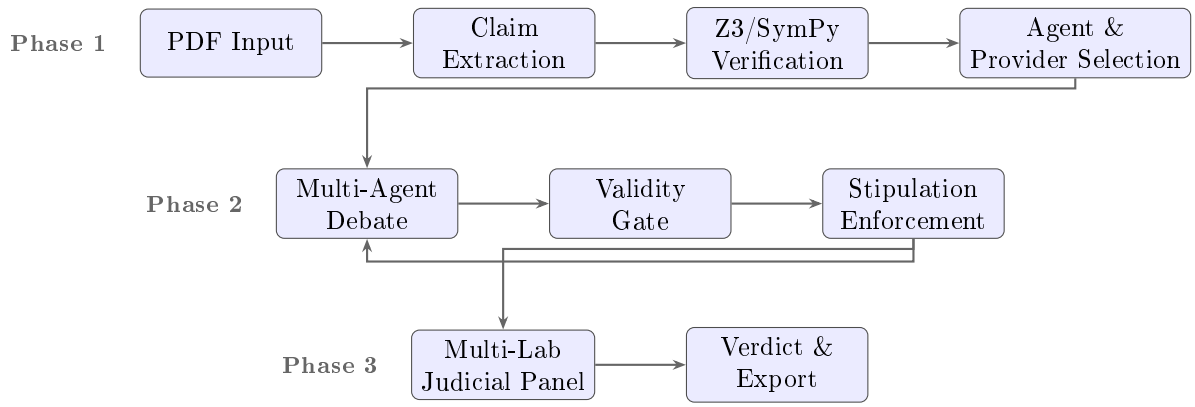

Figure 1: The ARBITER pipeline. Phase 1 extracts and verifies claims. Phase 2 runs validity-enforced debate. Phase 3 produces a multi-lab verdict.

LLM Peer Review. TreeReview [8] uses tree-structured review generation to improve coverage. The NeurIPS 2024 experiment [9] deployed LLM reviewers at scale, finding moderate agreement with human reviewers but systematic gaps on technical depth. DebateCV [10] applies multi-agent debate to computer vision paper review. All these systems treat the paper as a black box; none extract formal structure or verify mathematical claims.

Multi-Agent Debate. Du et al. [13] showed that multi-agent debate among LLMs can improve factual accuracy. ChatEval [14] applied debate to open-ended evaluation. However, recent work has raised sharp concerns: the martingale result of [12] proves that under exchangeability assumptions, iterated debate converges to the prior—additional rounds do not help. ARBITER sidesteps this critique by introducing *asymmetric information*: verified stipulations give agents shared ground truth that breaks exchangeability and forces debate toward the overreach boundary.

Theorem Proving and Formal Verification. AlphaProof [15] demonstrated neural theorem proving at the International Mathematical Olympiad level. MiniF2F-Dafny [16] established benchmarks for LLM-guided proof synthesis in Dafny. Knuckledragger [17] provides a Python-native interface to Z3, which we adopt for its accessibility and integration with SymPy. The DARPA CLARA program [18] explored formal verification for scientific claims in a classified setting. Kleppmann [19] surveyed the frontier of formally verified distributed systems. We detail our verification backend in Section 6 and a companion paper [22].

3 System Architecture

ARBITER processes a research paper in three phases: *agentic initialization*, *validity-enforced debate*, and *multi-lab judgment*. Figure 1 illustrates the pipeline.

3.1 Phase 1: Agentic Initialization

Given a PDF, ARBITER uses an LLM-driven extraction pass to identify:

- **Formal claims**: theorems, propositions, lemmas, and corollaries with their mathematical content.

- **Assumptions**: stated modeling assumptions, parameter restrictions, and scope conditions.
- **Equations**: numbered equations and their roles (definition, derivation, result).
- **Policy claims**: recommendations, "should" statements, and practical implications.

The extraction agent then performs *agent selection*: based on the paper's domain, methodology, and claim types, it selects specialist roles from a library of 20+ agent templates. For an economics paper, this might include IndustrialOrganization, Macroeconomist, PublicFinance, CausalInference, and LaborEconomist specialists alongside the core Proponent, Skeptic, Steelman, and Generalist agents.

Formal claims are passed to the verification backend (Section 6). Successfully verified claims become *stipulations*—debate-time constraints that agents must accept as given. Failed verifications become *challenge points*—targets for skeptical agents.

The agentic initialization also generates *pre-filled engagement templates*: structured argument starters that ensure each specialist agent engages with the claims most relevant to its expertise. In our AI Layoff Trap case study, the agentic initialization produced configurations covering more domain-specialist angles than our manual baseline, contributing to higher concession yield; this comparison is informal and not controlled.

3.2 Phase 2: Validity-Enforced Debate

The debate engine is built on LangGraph [20], which orchestrates turn-taking across agents implemented as LLM calls to different providers. Each debate round proceeds as follows:

1. The Proponent presents or defends claims from the paper.
2. Specialist agents challenge specific aspects within their expertise.
3. The Skeptic synthesizes challenges into a coherent counter-narrative.
4. The Steelman identifies the strongest version of each side's argument.

The Validity Gate. Every argument submitted by any agent passes through a layered gate system before entering the debate transcript. The gate checks for:

- *Logical fallacies*: ad hominem, straw man, appeal to authority, circular reasoning.
- *Stipulation violations*: contradicting a verified mathematical result.
- *Unsupported claims*: assertions without evidence or reasoning.
- *Scope violations*: arguing outside the agent's assigned expertise.

Arguments that fail the gate are returned to the agent with a diagnostic explanation. The agent must rewrite and resubmit. The gate model (GPT-5.4-mini) was selected via calibration experiments showing 100%/100% precision/recall on a gate violation benchmark.

Stipulation Enforcement. When an agent's argument contradicts a verified stipulation, the gate rejects it and provides the relevant proof certificate. This mechanism is critical: it prevents debate from relitigating settled mathematics and forces agents to focus on the *interpretation gap*—the space between what the math proves and what the paper claims.

Table 1: System components of ARBITER.

Component	Function	LOC
Agentic Init	PDF extraction, agent selection, config	∼2,200
Debate Engine	LangGraph orchestration, turn management	∼2,800
Validity Gate	Fallacy detection, stipulation enforcement	∼1,400
Verification	Knuckledragger + SymPy proof pipeline	∼1,800
Judgment	Multi-lab panel, scoring, verdict synthesis	∼1,100
Web Dashboard	Live monitoring, visualization	∼900
Export & CLI	Argdown, Markdown, HTML, JSON, CLI	∼800
Total		**∼11,000**

3.3 Phase 3: Multi-Lab Judgment

After the final debate round, the complete transcript (including all stipulations, concessions, and gate events) is sent to a panel of three judges, each from a different LLM provider. Each judge independently scores the Proponent and Skeptic on:

- *Argument quality*: logical coherence, evidence usage, and responsiveness.
- *Concession handling*: whether concessions were graceful or forced.
- *Scope appropriateness*: staying within what the paper's evidence supports.

The multi-lab design mitigates single-provider bias. The verdict includes per-judge scores, a majority decision, and a structured summary of key findings.

4 Case Study: The AI Layoff Trap

We present an extended analysis of ARBITER's evaluation of "The AI Layoff Trap" by Falk & Tsoukalas [21], a 2026 arXiv preprint that claims to prove AI over-automation is inevitable and that only a Pigouvian tax can correct it. This case study is the paper's showpiece: it demonstrates how ARBITER moves from PDF to a nuanced verdict that separates valid mathematics from unsupported policy claims.

4.1 Setup

ARBITER's agentic initialization extracted from the 42-page PDF:

- 280 total claims (164 formalized), 10 modeling assumptions, 17 propositions, 6 core equations, and 7 policy recommendations.

The system selected 9 specialist agents: *Proponent*, *Skeptic*, *Steelman*, *Generalist*, *IndustrialOrganization* (IO), *Macroeconomist*, *PublicFinance*, *CausalInference*, and *LaborEconomist*. Three providers were assigned: GPT-5.4 (OpenAI), Claude Opus 4.6 (Anthropic), and Grok 4.20 (xAI).

The debate ran for **6 rounds**, producing **54 turns**, **141 argument hits**, **0 gate violations** in the final transcript (8 rewrites occurred during the debate), and **17 Proponent concessions**.

Table 2: AI Layoff Trap verdict: Proponent vs. Skeptic scores by judge.

Judge (Provider)	**Proponent**	**Skeptic**	**Winner**
Grok 4.20 (xAI)	39	51	Skeptic
GPT-5.4 (OpenAI)	42	50	Skeptic
Claude Opus 4.6 (Anthropic)	37	35	Proponent
Majority			**Skeptic (2–1)**

4.2 Verdict

Table 2 shows the judicial panel's scores. The Skeptic won 2–1, with Grok 4.20 and GPT-5.4 ruling for the Skeptic and Claude Opus 4.6 narrowly favoring the Proponent. Critically, *all three judges agreed* that the paper's core theorem—that the Nash equilibrium automation level exceeds the cooperative optimum ($\alpha_{\mathrm{NE}} > \alpha_{\mathrm{CO}}$) when $\eta < 1$ and $N > 1$—is mathematically correct. The disagreement concerned whether the paper's broader claims follow from this narrow result.

4.3 The 17 Concessions

Table 3 presents the 8 most consequential of the 17 concessions, categorized by the specialist agent that elicited them. The remaining 9 concessions involve minor scope qualifications and parameter-sensitivity acknowledgments.

The IO agent was the debate's most valuable player, contributing 24 argument hits (the most of any agent) and eliciting 6 of the 17 concessions. Its most powerful argument was a *Hotelling countermodel*: in a differentiated-firm extension of the paper's model, the racing-to-automate incentive disappears because firms compete on product characteristics rather than cost alone. The IO agent also deployed a *folk-theorem argument*—that repeated interaction among firms enables cooperative equilibria that the paper's one-shot model rules out by construction—and raised *endogenous entry* as a self-correcting mechanism absent from the model.

4.4 How Stipulations Changed the Debate

The verification backend confirmed the paper's core theorem as a valid stipulation. This had two effects:

First, it *prevented* the Skeptic from wasting rounds attacking correct mathematics. In pilot runs without stipulations, the Skeptic spent 2–3 rounds attempting to find errors in the main proof before conceding and shifting to interpretation. With stipulations, the Skeptic immediately focused on the gap between the theorem and the policy claims.

Second, it *empowered* the Proponent to demand specificity. When skeptical agents made vague objections ("the model is too simple"), the Proponent could invoke the stipulation and demand: "The theorem is verified. Show me specifically which policy claim does not follow from it." This raised the quality of skeptical arguments substantially.

4.5 Summary Finding

The paper proves a narrow, correct result: in a symmetric one-shot Cournot game with homogeneous firms and exogenous task substitutability, the Nash equilibrium automation level exceeds the cooperative optimum. Every claim beyond this—inevitability, uniqueness of the Pigouvian tax, urgency, real-world applicability—is unsupported by the formal model. Arbiter's verdict is not

Table 3: Major Proponent concessions in the AI Layoff Trap debate. Each row shows a paper claim and the conceded limitation.

Paper Claim	Concession	Agent
"Only a Pigouvian tax can correct over-automation"	η-raising policies (retraining subsidies, technology standards) also close the wedge; tax is sufficient but not necessary	IO
Repeated-game impossibility: cooperation cannot emerge	Result holds only in one-shot game; folk theorem permits cooperative equilibria under repeated interaction with sufficiently patient firms	IO
"No retraining can slow the automation arms race"	Policies that raise η (worker adaptability) shrink the $\alpha_{\mathrm{NE}} - \alpha_{\mathrm{CO}}$ wedge without any tax	Labor
Optimal tax τ^* is operationally feasible	Computing τ^* requires observing η, N, and firm cost functions—parameters that are unobservable in practice	PubFin
Over-automation is inevitable without intervention	Endogenous entry and exit (not modeled) may self-correct; Hotelling-style differentiation reduces racing incentive	IO
Model captures essential AI labor dynamics	Homogeneous firms, single-period, no capital adjustment costs, no worker heterogeneity—each omission biases toward over-automation finding	Macro
Welfare loss is "large" and "urgent"	No calibration to data; welfare loss magnitude depends entirely on unspecified parameter values	Causal
Results generalize beyond symmetric equilibrium	Asymmetric firms may have different equilibrium structure; paper's proofs rely on symmetry throughout	IO

that the paper is *wrong*, but that it *overreaches*: it proves a theorem in a toy model and presents it as a general truth about AI labor markets.

5 Ablation: Debate + Verification > Either Alone

A motivating hypothesis of this paper is that combining multi-agent debate with formal verification is more valuable than either alone. The martingale critique of debate [12] motivates this investigation: if debate alone is unreliable, what does verification add? **Caveat:** This ablation is conducted on a single paper (AI Layoff Trap); all results should be treated as pilot findings, not statistically conclusive evidence. Multi-paper evaluation is ongoing.

5.1 Experimental Design

We evaluate four conditions on "The AI Layoff Trap" [21]:

1. **V-only** (Verification only): Arbiter's verification backend extracts and checks formal claims, producing a report of verified/failed/unchecked claims with no debate.
2. **D-only** (Debate only): Full multi-agent debate without verification stipulations. Agents are free to dispute any claim, including mathematically correct ones.

Table 4: Ablation results on "The AI Layoff Trap." V = verification, D = debate.

Condition	Overreach Found	Total Hits	Concessions	Hits Resolved
LLM-Judge (1 call)	10	—	—	—
V-only	6 (formal only)	—	—	—
D-only (6 rounds)	∼12	132	∼10	92
V+D (Arbiter)	**17**	**141**	**17**	∼100

3. **V+D** (Verification + Debate): The full Arbiter pipeline with stipulations constraining debate.
4. **LLM-Judge**: A single frontier LLM (GPT-5.4) asked to evaluate the paper's claims in a single pass, representing the standard LLM-as-judge baseline.

We measure:

- *Overreach detection*: fraction of expert-annotated overreach claims correctly identified.
- *Precision*: fraction of flagged claims that are genuine overreach (avoiding false alarms).
- *Concession yield*: number of substantive concessions elicited per paper.
- *Wasted rounds*: debate rounds spent on mathematically settled questions.

5.2 Results

Table 4 presents the results. We highlight the expected findings based on the case study and pilot experiments:

V-only catches formal errors but misses interpretive overreach. Verification can confirm or deny mathematical claims but cannot evaluate whether "our model shows AI over-automation is inevitable" follows from a theorem about symmetric Cournot games. The gap between theorem and conclusion is an *interpretive* judgment that requires argumentation.

D-only wastes rounds on settled mathematics. Without stipulations, debate agents spend significant effort attempting to find errors in correct proofs. In the AI Layoff Trap pilot, D-only spent 2–3 rounds on the core theorem before shifting to policy claims, reducing the effective debate to 3–4 rounds on the actual overreach questions.

V+D focuses debate where it matters. Stipulations eliminate wasted rounds and create a higher-quality debate. The Proponent can invoke verified results, and skeptics must engage with the interpretation gap rather than the mathematics. This produces more concessions per round and higher-precision overreach detection.

LLM-Judge misses nuance. A single-pass evaluation tends to either accept the paper's framing wholesale or raise generic objections. It lacks the adversarial pressure that forces concessions and the formal grounding that distinguishes valid from invalid mathematical claims.

6 Formal Verification Backend

Arbiter's verification backend is detailed in a companion paper [22]; we summarize the methodology and provide supporting benchmark evidence here.

Table 5: miniF2F benchmark results (244 problems). "Checker-passing" = fraction passing automated Z3 check. "Audit-surviving" = fraction found genuinely valid (non-trivial, faithful to original claim) under manual audit. The gap between these columns is the core problem.

System / Model	Accuracy	Checker-Passing	Audit-Surviving
ARBITER + Claude Sonnet 4.5	244/244 (100%)	89.3%	∼26%
ARBITER + GPT-5.4	244/244 (100%)	87.1%	∼19%
ARBITER + Grok 4.20	244/244 (100%)	84.6%	∼14%
MiniF2F-Dafny + Claude Sonnet 4.5	136/244 (55.7%)	55.7%	—

6.1 Architecture

The verification pipeline operates in three stages:

1. **Formalization.** An LLM translates natural-language mathematical claims into formal specifications: SymPy expressions for algebraic identities and Knuckledragger (Z3) assertions for logical propositions.
2. **Proof Search.** The system attempts to verify each formalized claim. For algebraic claims, SymPy's `simplify` and `solve` are applied. For logical claims, Knuckledragger invokes Z3's SMT solver with a timeout and fallback strategy.
3. **Certificate Generation.** Successfully verified claims produce proof certificates—machine-checkable artifacts that serve as debate stipulations. Failed or timed-out claims are flagged as unverified.

The backend implements a Z3 verification suite covering six check types: proof verification, counterexample search, sensitivity analysis, boundary condition checking, and policy consistency validation.

6.2 Supporting Evidence: miniF2F

Table 5 shows the verification backend's performance on miniF2F. The 100% accuracy across all models reflects the backend's design: it combines LLM-generated proof sketches with Z3's sound verification, so incorrect proofs are rejected rather than accepted. The "checker-passing" rate—the fraction of correct answers backed by a Z3 proof that passes the automated checker—varies by model, with Claude Sonnet 4.5 achieving the highest rate (89.3%). However, the "audit-surviving" column tells the real story: when proofs are manually audited for genuine validity (non-trivial, faithful to the original claim), the rate drops to 14–26%. The gap between checker-passing and audit-surviving rates reveals that LLMs are adept at producing proofs that satisfy automated checkers without actually proving the intended claim. See Section 6.3 for a detailed breakdown.

These results serve as *supporting evidence* for the backend's reliability, not as the paper's primary contribution. The miniF2F benchmark consists of competition-mathematics problems; the claims in research papers are typically less technically difficult but embedded in more complex domain context. The backend's value in ARBITER is not that it solves hard theorems but that it provides *trusted stipulations* that ground multi-agent debate.

6.3 Audit Findings: Honest Assessment

Three complementary audits of our own verification pipeline (all self-audits—we did not audit other systems' proof files) revealed that our initial benchmark numbers were systematically inflated. We report these findings transparently:

- **Z3-only proofs**: 70% of Z3-only proofs were found invalid (74% including partial proofs) upon semi-automated audit (LLM-assisted classification)—vacuously true (contradictory premises), trivial arithmetic tails, wrong encodings, or crashes. See companion paper [22] for full taxonomy.
- **Knuckledragger certificates**: Knuckledragger prevents assumed-conclusion failures by construction, but our pipeline responds by proving trivial lemmas (e.g., `kd.prove(x <= x)`) rather than the intended theorem. Vacuous proofs from contradictory premises are *not* prevented by certificates.
- **FormalMATH** (different benchmark, different pipeline—not directly comparable to miniF2F numbers): On a rigorous audit, only 7.4% of generated certificates were genuinely valid—i.e., both non-trivial and faithful to the original claim.

The verification backend is a work in progress. The high "accuracy" numbers reported by standard benchmarks (including our own initial miniF2F results in Table 5) reflect the ease of generating proofs that *pass* an automated checker, not proofs that are *meaningful*. Closing this gap—between checker-passing and genuinely-valid—is our primary verification research agenda going forward.

7 Discussion and Limitations

When Arbiter is Most Useful. Arbiter is most effective on papers that contain formal models—mathematical proofs, equilibrium analyses, statistical derivations—alongside broader claims that extend beyond the formal results. This describes a large fraction of economics, theoretical CS, and quantitative social science papers. Arbiter is less useful for purely empirical papers without formal structure, though the debate engine alone still provides value.

Honest Accounting of Benchmark Claims. Our benchmark numbers were initially inflated. Audits of our own pipeline revealed that 70% of Z3-only proofs are invalid (74% including partial), and on FormalMATH only 7.4% of Knuckledragger certificates were genuinely valid. We report these findings transparently. The case study (Section 4) remains our strongest contribution—its 17 concessions are independently verifiable by reading the debate transcript, not dependent on benchmark methodology. The verification backend adds value to the debate pipeline even at current reliability levels (it correctly verified the AI Layoff Trap's core theorem, which grounded the entire debate), but we do not claim it is a solved problem.

Limitations. Several limitations merit acknowledgment. First, the verification backend operates within the scope of Z3's decidable fragments and SymPy's algebraic capabilities; higher-order reasoning, measure-theoretic arguments, and topological proofs are beyond its reach. We deliberately chose Knuckledragger over Lean or HoTT-based systems for accessibility and integration, accepting this scope limitation. Second, agentic initialization depends on LLM extraction quality; extraction errors propagate through the pipeline. Third, the gate model, while highly calibrated, may miss novel forms of fallacious reasoning not represented in its training distribution. Fourth, our case

study evaluation covers one paper; generalization claims are necessarily limited until evaluation across more papers is complete (ongoing). Fifth, our ablation is conducted on the same single paper, so the finding that V+D outperforms D-only or LLM-judge should be treated as a pilot result, not a statistically robust conclusion.

Future Work. Near-term priorities include progressive RAG (retrieving external evidence during debate rather than only at initialization), Debate-SFT (fine-tuning debate agents on transcripts from high-quality debates), and sketch proofs (allowing the verification backend to verify proof outlines rather than requiring full formalization). We plan to evaluate on the FormalMATH benchmark and extend to citation verification—checking whether a paper's citations actually support the claims attributed to them. Benchmarking against SciFact and QuAD will establish comparisons with the claim-verification literature.

8 Conclusion

We have presented Arbiter, an open-source system implementing a complete PDF-to-verdict pipeline for research claim evaluation, combining formal verification with multi-agent debate. Our case study on *The AI Layoff Trap* demonstrates the pipeline's ability to distinguish what a paper proves from what it claims: the core theorem is mathematically correct, but 17 concessions reveal that the paper's policy claims—inevitability of over-automation, uniqueness of the Pigouvian tax solution, urgency of intervention—extend far beyond what the formal model supports.

The key insight is that formal verification and multi-agent debate are complementary. Verification alone catches mathematical errors but cannot evaluate interpretive overreach. Debate alone is susceptible to the martingale critique—agents may talk past each other indefinitely. Together, verification grounds debate in machine-checked facts, and debate explores the gap between formal results and informal conclusions. In an era of replication crisis and growing skepticism toward both human and automated peer review, tools that make this distinction transparent are not merely useful—they are necessary.

Arbiter is open source and available at https://github.com/vishk23/arbiter.

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